RAVEL: A Spatial Coding Harness for Reasoning about Change and Preservation

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Abstract

A physical action can change one relation while preserving another involving the same moving object. We study selective spatial relation reasoning: jointly establishing relation endpoints and distinguishing change from preservation within and outside an action's participation scope. Our dataset contains 12,000 indoor questions, including a 4,000-question test set spanning observed updates, unseen consequences, constrained edits, and consecutive actions. We introduce Relation-Aware Validity and Evidence Linking (RAVEL), a training-free harness that models scenes, actions, relation dependencies, and evidence as a structured spatial representation in code. The harness executes its semantics to update valid attributes, withdraw invalid premises, and combine supported computations with a frozen vision-language model's judgments into joint answers. The complete configuration improves whole-question accuracy by 13.35–19.80 percentage points across five backends, using additional initial geometry and statewise calls. Matching raw information and call schedules on 500 questions retains gains of 18.20 points for Qwen3-VL-4B and 2.20 points for Astra. Controlled interventions locate Qwen's largest increment in executable organization, with smaller, backend-dependent binding effects. Joint relation scoring shows improvements in both change and preservation, while trajectory and provenance analyses expose unresolved visual judgments and retained initial mistakes. These results connect the validity of relational premises to the quality of complete answers under physical actions.

1. Introduction

Understanding an action requires reasoning about the relations it changes together with those it preserves. In Figure 1, releasing A makes it collide with B and causes B to move. A–B height and contact change, B remains on the floor, and fixed C–D direction persists. The same motion of B thus accompanies both change and preservation. A correct account must establish each relation's endpoints and distinguish preservation after participation from preservation outside the interaction's scope. We call this *selective spatial relation reasoning*.

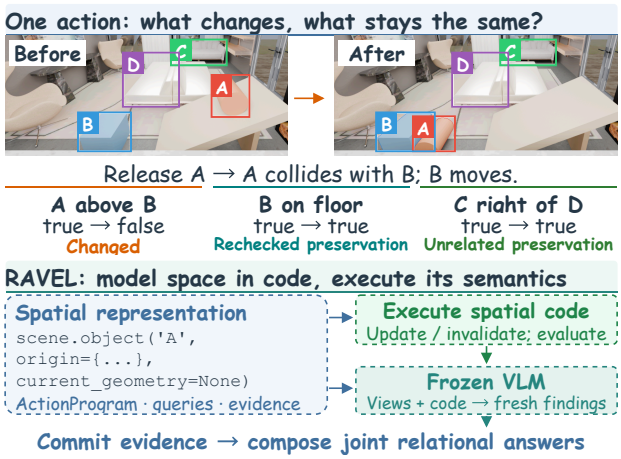


Figure 1. **One action, different relational consequences.** B moves yet remains supported; fixed C–D direction also persists. RAVEL models the scene, actions, dependencies, and evidence in code (fields abridged). Its harness executes attribute and query semantics, obtains unresolved findings from a frozen VLM, and composes joint answers. Both Astra methods solve this recorded example. Crops are identical; image access varies by task.

Existing work tests spatial estimation, dynamic memory, and physical prediction [2–4, 8, 12, 13]. We bring relation truth and action participation into a joint evaluation: detecting a collision is insufficient if the same answer loses a preserved support relation. Our benchmark spans observed transitions, unseen outcomes, multi-relation updates, constrained edit selection, and consecutive actions, with 12,000 questions and a 4,000-question test set. Preservation includes false relations: an edit that achieves its goal but violates a required unchanged relation is inadmissible.

The central inference problem is deciding which premises still establish a relation after an action. A prescribed translation can update a known center and support several direction or proximity judgments. Free motion invalidates that center as evidence of the final pose, even though the declaration remains in history. Meanwhile, fixed references can retain valid geometry and prior observations.

Participation alone cannot resolve these judgments: a moving object can preserve support while changing its distance to another object. The solver needs current endpoint evidence before interpreting change and preservation.

We introduce Relation-Aware Validity and Evidence Linking (RAVEL), a harness built around a structured spatial representation in code. Scene declarations encode object origins, reference frames, and valid attributes; action programs define how these attributes evolve; queries declare their dependencies and associated evidence. The harness interprets this representation to propagate or invalidate premises, compute supported relations, and ask a frozen vision-language model for unresolved findings. It commits these findings and composes joint answers about change, preservation, or admissible edits. The representation supplies both model context and harness execution semantics.

Across five backends, the full configuration improves whole-question accuracy by 13.35–19.80 points. Since it adds public initial geometry and state-wise calls, we match both factors on 500 questions: Qwen improves from 6.20% to 24.40%, and Astra from 78.00% to 80.20%. Binding and geometry interventions distinguish the organization bundle from individual enforcement rules. Relation-level gains extend to both change and preservation for Qwen; complete trajectories remain difficult, and provenance reveals both repeated fresh errors and reused initial mistakes.

Our contributions are a benchmark that jointly tests relation truth and action participation; a spatial coding harness linking executable declarations to evidence and joint decisions; and an evaluation combining five-backend comparisons, matched interventions, and relation-level diagnosis.

2. Related Work

Spatial and physical reasoning. SpatialVLM [3] develops large-scale spatial question answering; DSI-Bench [13] decouples observer and object motion; SpaMEM [8] studies action-conditioned spatial belief evolution. CLEVRER [12], Physion [2], and PhysBench [4] evaluate complementary aspects of physical events and prediction (Tab. 1). Our benchmark couples relation endpoints with execution participation, separating two forms of preservation across five task interfaces. HSSD [7] and Infinigen Indoors [9] supply the indoor scene sources.

Executable reasoning and harnesses. VISPROG [5] and ViperGPT [11] compose executable visual modules. CROSS [10] uses typed geometric operators for video spatial reasoning. VISTA [6] retains visual history and lets a model actively revisit images through a visual harness. RAVEL encodes scenes, actions, and relational evidence in a structured spatial representation with explicit execution semantics. Its harness interprets attribute validity and query dependencies, then composes computed and model-provided findings under a fixed call schedule. The representation is compiled from public declarations; the model supplies observations rather than generating programs or selecting tools.

Table 1. **Evaluation targets and inference mechanisms.** Top: benchmark scope. Bottom: how evidence enters executable reasoning. Entries summarize documented mechanisms.

A. Benchmarks	**Evaluation target**
SpatialVLM [3]	Qualitative and metric spatial questions
DSI-Bench [13]	Observer and object motion
SpaMEM [8]	Action-conditioned spatial memory
CLEVRER [12]	Descriptive, explanatory, predictive, and counterfactual collision reasoning
Physion [2]	Prediction across physical phenomena
PhysBench [4]	Physical properties, relations, scenes, dynamics
Our benchmark	Joint endpoints and participation; change, two preservation types, edits, consecutive updates
B. Harnesses	**Execution and evidence mechanism**
VISPROG [5]	Generated programs compose visual modules
ViperGPT [11]	Python programs combine vision APIs
CROSS [10]	Typed geometric operators and spatial skills
VISTA [6]	Active image revisiting; raw visual history
Direct VLM	One call; final-answer schema
RAVEL harness	Spatial code; action-dependent attribute validity; typed evidence; joint composition

3. Benchmark: Selective Relation Reasoning

3.1. Relation Truth and Participation

An episode pairs states $s_0, \ldots, s_{L-1}$ with actions $a_0, \ldots, a_{L-2}$ and queries $\mathcal{Q}$. Objects $\{A, B, C, D\}$ have persistent red/blue/green/purple identities and include controlled, free, and fixed participants. Each state has two views, subject to task permissions. Relation values $v_{t,q} \in \{\text{true}, \text{false}\} \cup \{\bot\}$ are true, false, or unknown at a threshold boundary or under insufficient evidence.

The seven predicates in Figure 2 use center, surface, interior, or joint geometry. Distances use meters in the fixed initial-view-1 ground frame, with vertical z. Direction and proximity use centers; `on_surface` denotes bottom contact or proximity to a named upward surface, not force equilibrium (Appendix B).

Let $\mathcal{D}(q)$ contain the query's named objects and $\mathcal{A}_t$ the directly controlled or secondarily moving participants. An update combines endpoint truth and participation:

$$u_{t,q} = \begin{cases} \textsf{unknown}, & \bot \in \{v_{t,q}, v_{t+1,q}\}, \\ \textsf{changed}, & v_{t,q} \neq v_{t+1,q}, \\ \textsf{rechecked_preserved}, & \mathcal{D}(q) \cap \mathcal{A}_t \neq \varnothing, \\ \textsf{unrelated_preserved}, & \text{otherwise.} \end{cases} \tag{1}$$

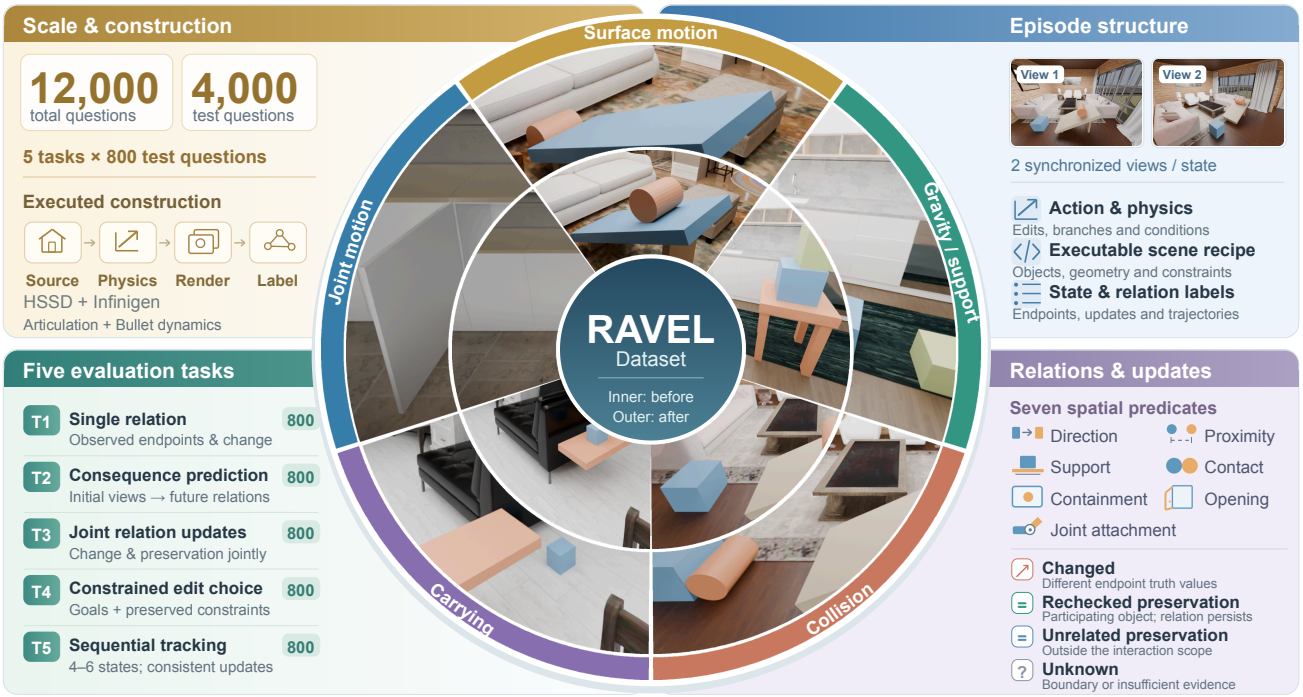


Figure 2. **RAVEL dataset overview.** The dataset contains 12,000 questions, including 4,000 test questions with 800 per task. Paired scene views illustrate five physical categories (inner: before; outer: after). Construction, episode structure, task interfaces, and relation labels define the benchmark. Image permissions depend on the task (Tab. 6).

The last two cases require known, equal values, including false-to-false preservation. Thus B–floor can remain true while B moves, making it rechecked; fixed C–D direction is unrelated. Gold participation describes recorded execution scope, rather than counterfactual causal effects.

3.2. Tasks, Construction, and Evidence Access

The dataset contains 12,000 questions from HSSD and Infinigen scenes; the test set has 4,000 questions, 800 per task, from 100 source groups. Tasks cover joint motion, gravity/-support, surface motion, collision, and carrying. Groups share layouts or assets and define statistical dependence.

T1/T3 score observed endpoints and three-/four-way updates for single/multiple relations; T1 merges preservation types. T2 predicts future relations from initial views and actions. T4 selects all independent edits satisfying every goal and preservation constraint. T5 tracks endpoints and updates across four to six states. T2 futures and T4 candidate outcomes remain hidden; other tasks expose the relevant states (Tab. 6).

Blender executes prescribed articulation and Bullet dynamics, records states and interactions, and labels predicates and participation. Each T4 candidate resets to the same initial state. Automated acceptance checks execution, identity, physics, and label/input consistency independently of model answers. Inputs contain permitted images, actions, declarations, queries, and schemas; hidden outcomes and mesh-level scoring evidence remain private.

The test set contributes 77,303 answer items, including 27,664 T3/T5 updates; 76.5% of these updates are preservation. Full label counts, execution settings, and the two unresolved annotations appear in the supplement (Tab. 15 and Appendix D).

4. RAVEL: A Spatial Coding Harness

4.1. Model Space in Code

RAVEL models spatial reasoning through executable declarations (Figure 3). From a public initial-construction recipe and declared actions, the harness builds `SceneSpec`, `ActionProgram`, queries, history, and evidence. `SceneSpec` separates object origins and initial geometry from currently valid attributes in a shared reference frame. Queries name predicates and their attribute dependencies: A–B height reads centers; A–B contact reads surfaces.

The harness interprets action programs to reconstruct current attributes from the recipe and action history. Fixed objects retain geometry; a prescribed pure translation updates a valid center as $c_{t+1}(o) = c_t(o) + \Delta_t$. Rotation and free evolution of unheld objects invalidate their centers, leaving initial geometry historical and `current_geometry` absent. Joint operations update de-

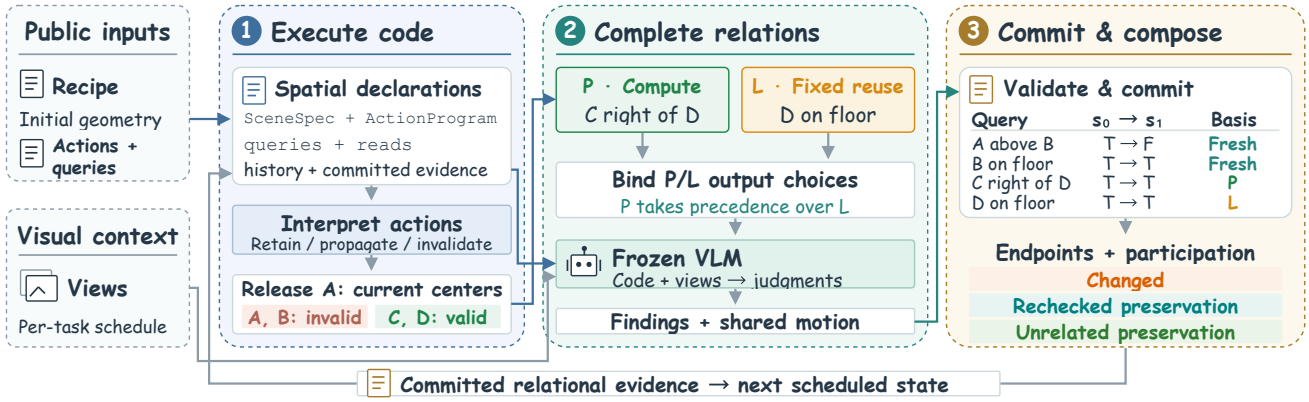


Figure 3. **RAVEL execution loop.** **(1)** Public recipes, actions, and queries become spatial code; interpreting actions reconstructs valid attributes. **(2)** Program findings (P) and eligible fixed evidence (L) bind typed output choices. The frozen VLM reads code and permitted views to complete relations and shared motion. **(3)** The harness validates and commits findings, then composes endpoint values and participation into joint answers. Committed relational evidence carries forward to the next scheduled state. The example follows the Astra collision trace in Tab. 7; T/F denote true/false.

cleared opening coordinates. Python declarations expose this state to the frozen model; the interpreter does not simulate free-body trajectories. Initial centers are authored AABB centers transformed into the question frame, supplied additionally in the main comparison and equally to matched Direct.

4.2. Evaluate Relational Queries

Program findings (P). Direction and proximity execute when both current centers are valid and stored error bounds permit a threshold decision. Declared or prescribed joint coordinates establish opening; authored joint topology establishes attachment. Program findings bind the structured output and take precedence over retained model evidence. After A's release, C–D direction remains computable; A and B lose authoritative centers.

Fixed model evidence (L). A known finding is retained when all named operands are fixed. It retains its original state and evidence basis, as with D–floor support in the collision example. Fixed dependencies establish continued applicability, but an incorrect original finding remains incorrect when reused.

Fresh model judgments. The model resolves remaining relations using permitted images, spatial code, and predicate-specific vocabularies (Tab. 8). Contact, support, and containment require surface or interior evidence beyond centers and bounding boxes. The harness projects findings to true, false, or null. For participation, direct targets are controlled and fixed objects still; the model judges secondary motion of other objects using a 0.025 m center-displacement threshold. Shared participant status then enters every relevant update.

4.3. Compose Joint Decisions

For T3/T5, committed endpoint values and participation determine updates by Equation (1). In the collision trace, A–B height/contact change, B–floor support is rechecked preservation, and fixed C–D direction is unrelated preservation. T1 merges both preservation classes. For T4, let $\mathcal{G}$ contain goal query–value pairs, $\mathcal{K}$ preservation queries, and $\hat{v}_{c,q}$ the prediction under independent candidate c:

$$
\begin{aligned}
G(c) &= \bigwedge_{(q,b)\in\mathcal{G}} [\hat{v}_{c,q} = b], \\
K(c) &= \bigwedge_{q\in\mathcal{K}} [(\hat{v}_{0,q} \neq \bot) \wedge (\hat{v}_{c,q} = \hat{v}_{0,q})], \\
\widehat{\mathcal{C}} &= \{c \in \mathcal{C} : G(c) \wedge K(c)\}.
\end{aligned}
\tag{2}
$$

Every accepted edit achieves all goals and preserves every specified known truth, including false values. Composition enforces these rules over predictions; wrong findings can still yield wrong answers.

4.4. Execute, Complete, and Commit

Each call exposes the spatial code and binds already established P/L findings to singleton output choices; unresolved queries retain their typed vocabularies. This interface carries the representation's execution results into model decoding. The frozen VLM returns relation observations and shared motion judgments; the harness validates, commits, and projects them into joint answers. The model updates relational evidence, not numerical 3D geometry. T1/T3 call once per endpoint; T5 calls once per observed state.

Table 2. **Full-cohort results (4,000 questions; 77,303 answer items).** Metrics are percentages; ΔWhole is RAVEL minus Direct in points. Bold marks the better Whole, Items, T5, and F1 within each backend. Joint accuracy requires both endpoints and the T3/T5 update. RAVEL adds declarations and uses 8,995 calls versus 4,000; matched controls follow in Tab. 3.

		Overall			Whole by task					Changed	Joint endpoints + update		
Backend	Method	Whole	Items	ΔWhole	T1	T2	T3	T4	T5	F1	Changed	Recheck.	Unrel.
Qwen3-VL-4B	Direct	3.60	44.66	–	14.38	2.25	0.00	1.83	0.00	18.85	5.18	16.94	38.50
	RAVEL	**23.40**	**78.92**	+19.80	67.75	10.50	12.60	23.17	**5.63**	**62.40**	50.36	62.68	87.98
Spatial-SSRL-4B	Direct	1.78	37.59	–	8.00	0.13	0.00	1.00	0.00	12.14	2.04	7.96	39.52
	RAVEL	**18.88**	**75.74**	+17.10	62.00	6.75	7.30	17.67	**3.25**	**56.06**	43.28	57.56	87.39
Molmo2-ER	Direct	8.75	47.45	–	39.50	2.75	0.00	2.00	0.00	38.03	0.49	39.84	0.00
	RAVEL	**22.10**	**74.67**	+13.35	65.50	11.13	9.20	22.50	**5.50**	**53.13**	35.33	62.34	78.84
Sol	Direct	62.05	92.76	–	88.13	55.38	55.20	64.00	49.75	88.13	81.63	93.08	94.84
	RAVEL	**79.00**	**96.64**	+16.95	96.00	69.00	77.80	80.83	**72.13**	**95.49**	93.43	96.81	97.64
Astra	Direct	62.70	92.83	–	87.50	57.38	54.30	66.50	50.88	88.25	81.60	93.73	94.30
	RAVEL	**81.20**	**96.96**	+18.50	95.75	73.75	79.10	86.50	**72.75**	**95.71**	94.30	96.78	97.54

Observed-state calls use current and adjacent previous two-view images; earlier actions and findings persist in the code context. T2 advances the context symbolically and makes one initial-view call; T4 makes one initial-view call over four independently initialized branches. Free-body outcomes remain model predictions after numerical premises are invalidated. No hidden endpoints or gold participation enter inference. The same rules apply to all backends; Appendices B.2 and C and Tab. 7 give pseudocode, actual declarations and requests, and the expanded trace.

5. Experiments

5.1. Protocol and Full-Configuration Results

We evaluate on the 4,000-question test set, with 800 questions per task. We evaluate Qwen3-VL-4B-Instruct [1], Spatial-SSRL-4B, Molmo2-ER, and API backends Sol (`gpt-6.1-sol`) and Astra (`gpt-6-astra`), both at medium reasoning effort. Direct receives the question, permitted images, common semantics, and final schema in one call. RAVEL additionally receives public initial geometry and declarations and uses state-wise inference. Both use constrained structured output without answer repair. All 40,000 main answers have valid format; differences concern semantic correctness.

Whole-question accuracy requires every semantic item to be correct; item accuracy micro-averages individual decisions. T4 Whole requires the exact admissible set. For T3/T5, changed F1 scores change labels, while joint accuracy requires both endpoints and the update label. We also decompose T5 into state-vector and complete-trajectory correctness. Paired 95% bootstrap intervals resample source groups for the fixed recorded runs; they do not estimate repeated-decoding variability (Appendix A.3).

The full configuration improves Whole by 13.35–19.80 points across all five backends (Tab. 2). Local models reach 18.88–23.40% Whole and 74.67–78.92% Items; Sol and Astra reach 79.00% and 81.20% Whole. Astra corrects 851 Direct errors and regresses on 111 questions, with a net gain interval of $[16.29, 20.78]$ points. This comparison measures the combined effect of additional declarations, state-wise calls, and executable organization. We next match access to examine the latter more directly.

5.2. Organization under Matched Access

Balanced500 samples metadata across all 100 source groups, with 100 questions per task and 20 per task–physics cell. Each control uses identical raw initial information, legal images, 1,100 state-scheduled calls, and output caps of 1,536 tokens for Qwen and 8,192 for Astra. Matched Direct predicts semantic decisions with a controller that stores and assembles them. Qwen's full-RAVEL reference reuses request-matched frozen responses; Astra's is rerun contemporaneously. Development examined 1,000 main questions and exposed 99 source groups; balanced500 contains 140 examined questions (Appendices A.2 and D.1).

Under this match, Qwen improves from 6.20% to 24.40% Whole, a gain of 18.20 points with interval $[13.98, 22.55]$. Astra improves from 78.00% to 80.20%, a 2.20-point gain with interval $[0.00, 4.57]$, while changed F1 decreases from 95.52% to 94.57%. Equal raw access and calls therefore leave a substantial Qwen benefit and a smaller, mixed Astra outcome (Tab. 3 and Figure 4). Actual tokens and time also differ (Sec. 6.4).

Compulsory binding. We cross program binding P and the fixed-evidence lock L: C00 disables both, C10 enables P, C01 enables L, and C11 is full RAVEL. All cells retain computed hints, typed findings, earlier evidence, and deterministic projection. Qwen gains 15.40 points from

Table 3. **Matched controls on balanced500.** Each row uses 1,100 calls. P/L enforce program findings/fixed evidence; C00–C11 retain hints and history. Cons. invalidates non-fixed centers after every action. Bold: column best. Colored Δ: signed point differences, not significance; brackets give paired 95% intervals. Contrasts are rounded separately.

			Qwen3-VL-4B			Astra		
Configuration	P	L	Whole	Items	F1	Whole	Items	F1
Direct	–	–	6.20	52.64	23.65	78.00	96.14	95.52
C00	0	0	21.60	78.56	62.15	80.00	**96.73**	**95.53**
C10	1	0	23.20	79.21	63.82	78.40	96.53	94.66
C01	0	1	21.40	78.86	62.64	79.40	96.59	95.03
C11: RAVEL	1	1	**24.40**	**80.03**	**65.50**	**80.20**	96.59	94.57
Cons. geometry	1	1	20.40	–	57.45	79.20	–	94.71
ΔWhole (C11−Direct)			+18.20 [13.98, 22.55]			+2.20 [0.00, 4.57]		
ΔF1 (C11−Direct)			+41.86 [35.25, 47.93]			-0.94 [-2.64, 0.55]		

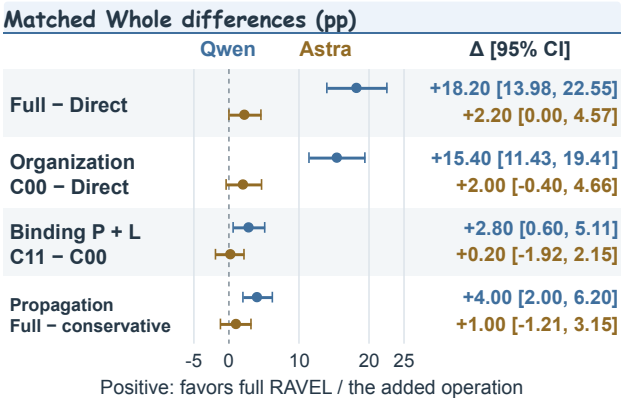


Figure 4. **Organization, binding, and propagation.** Paired Whole differences with source-group bootstrap 95% intervals on balanced500. Qwen is blue; Astra orange. C00 retains computed hints and history. The organization contrast jointly changes several mechanisms; all conditional contrasts are in Tab. 23.

matched Direct to C00, then 2.80 $[0.60, 5.11]$ from C00 to C11. The first contrast changes an organization bundle and does not isolate a single component. With L active, adding P gives 3.00 points; without P, adding L gives -0.20. Astra's C00/C11 scores are 80.00/80.20, with binding-effect intervals spanning zero. When P is disabled, Astra follows all 2,169 hints in both cells; Qwen overrides 26/25. Forced binding thus adds little when a backend already follows the available measurements.

Geometry propagation. Conservative invalidation discards all non-fixed centers after every action, including legal pure translations. Restoring selective propagation raises Qwen Whole from 20.40% to 24.40%, a 4.00-point difference $[2.00, 6.20]$, and changed F1 from 57.45% to 65.50%. Astra gains 1.00 point $[-1.21, 3.15]$, with a small F1 decrease. Retaining action-supported geometry helps Qwen compared with returning those judgments to the model; the marginal effect remains backend-dependent. Complete conditional contrasts appear in Tabs. 23 and 24.

6. Analyzing Joint Relation Judgments

6.1. Changes, Preservation, and Complete Trajectories

The joint scores in Tab. 2 connect complete-answer gains to relation-level judgments. Qwen's rechecked/unrelated joint accuracy rises from 16.94/38.50% to 62.68/87.98% on the full cohort. On matched balanced500, changed/rechecked/unrelated joint accuracy rises from 2.29/31.63/19.20% to 52.00/67.27/88.53%. False change alarms fall from 21.04% to 5.26% for rechecked relations and from 14.58% to 0.24% for unrelated ones, although changed recall remains 53.57%. Thus Qwen improves both detecting change and retaining correct preserved values. Astra's matched joint scores remain nearly unchanged (Tabs. 29 and 30). Aggregate gains also conceal trade-offs: full-cohort Molmo changed recall falls from 39.91% to 38.59% as precision rises from 36.32% to 85.24%.

Complete T5 answers remain harder than individual decisions. On 800 full-cohort trajectories, RAVEL reaches 72.13/72.75% Whole for Sol/Astra and 3.25–5.63% for local models. Astra gets the entire initial vector correct in 728 trajectories, but only 582 have every state and update correct. On 100 matched T5 trajectories, Qwen improves initial-vector accuracy from 3% to 60%; with RAVEL, nine have every state vector correct, twelve every update correct, and eight both. Astra reaches 97.29% Items but 71% Whole, versus 70% matched Direct. These are joint-correctness measurements; identifying retained errors requires provenance, not accuracy alone (Tabs. 16 and 26).

6.2. Validity and Correctness of Evidence

Each full Qwen/Astra run records 55,439 findings, of which 46,439 align with relation-value annotations; 9,000 T4 branch findings lack that alignment. Fresh model findings are correct in 64.51/95.08% of scorable instances for Qwen/Astra, and fixed reuse in 83.44/95.94%. All 14,881 scorable program findings per backend agree with annotations. These source-specific rates cover different queries and states (Figure 5a). Qwen records 917 wrong reuses from 428 initial sources; Astra records 224 from 114. Fresh judgments also yield 9,236/1,281 errors, including 3,118/394 transitions from a previously correct finding (Tab. 35). Continued applicability and initial correctness are distinct properties of evidence.

Retention horizon. On the 100 matched T5 trajectories, we cap original-source age at $h \in \{0, 1, 2, \infty\}$ in both persistent and previous-state channels; copying does not reset

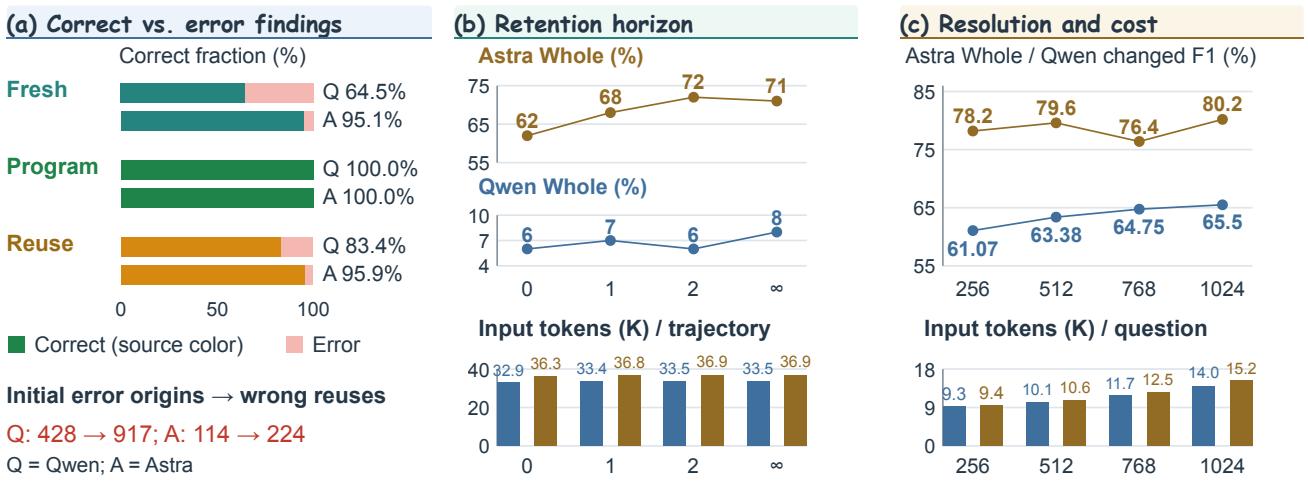


Figure 5. **Evidence quality and sensitivity.** (a) Correct/error fractions among scorable full-run findings; Qwen/Astra counts differ for fresh (26,022/26,039) and reused findings (5,536/5,519), with 14,881 program findings each. (b) T5 Whole versus source-age cap on 100 matched trajectories. (c) RAVEL resolution on balanced500. Blue: Qwen; orange: Astra. Quality curves use local ranges; token bars start at zero. Curves show point estimates; paired-difference intervals appear in Tabs. 25 and 27.

age. Images, actions, initial declarations, and numerical evaluation remain available. Qwen Whole is 6/7/6/8%; Astra is 62/68/72/71% (Figure 5b). Astra's $h = 0$ difference from default is -9.00 points $[-20.18, 2.22]$. Longer retention has no monotonic advantage in these runs; complete intervals and costs appear in Tab. 25.

Recorded traces. In the translation case, Astra Direct keeps A–C proximity true throughout; RAVEL correctly tracks its alternating values with valid numerical geometry (Figure 6a). In the tray-rotation case, Direct is correct but RAVEL repeatedly misses B–floor support after rotation (Figure 6b). These are fresh judgments about a moving operand, so the error is not fixed-evidence reuse. A separate Qwen support trace (R5) retains an error: an incorrect initial finding for fixed D is reused at states 1–3; an Astra trace (R1) provides another instance. The records distinguish successful propagation, repeated fresh misjudgment, and retained initial error.

6.3. Visual and Physical Sensitivity

Unseen physical outcomes remain difficult: local RAVEL backends reach only 6.75–11.13% T2 Whole, versus 69.00–73.75% for API backends. Matched Qwen improves direction-value accuracy from 46.49% to 85.68%, while support changes from 70.44% to 69.69%; its collision Whole is 9%. Astra's matched joint-motion Whole falls from 59% to 57%, with slight decreases in contact and proximity. Full task, physics, predicate, participation-role, and source breakdowns appear in Appendices F and G.

Resolution. Resizing permitted images to 256/512/768/1,024 pixels preserves aspect ratio and all other RAVEL settings. Qwen Whole stays within 23.00–24.60%, but changed F1 falls from 65.50% at 1,024 to 61.07% at 256. Astra Whole is non-monotonic: 76.40% at 768 versus 79.60% at 512. Lower resolution reduces input tokens while affecting metrics differently (Figure 5c,Tab. 27). Direct is not rerun at these resolutions.

Appearance and sources. We evaluate both matched methods under one alternative lighting and one material configuration on balanced500, keeping physical states, cameras, labels, and boxes fixed. Qwen's RAVEL Whole is 24.00/24.20%, with 19.00-point gains in both conditions. Astra gains 1.40 $[-1.40, 4.24]$ and 2.80 $[0.39, 5.38]$ points (Figure 7a). Paired changes in the gain relative to original images have intervals spanning zero. Both source families also improve in the full comparison: Astra gains 17.80 points on HSSD and 22.19 on Infinigen. The source slices overlap development; appearance tests isolate rerendering of existing episodes (Tabs. 12 and 28).

6.4. Accuracy and Inference Cost

The full configuration uses 8,995 calls per backend versus 4,000 for Direct. For Astra, mean input tokens increase from 5,626 to 15,460 and cumulative inference time from 22.18 to 33.92 seconds per question (Figure 7b). Time sums recorded calls within each question; hardware and API conditions differ across backends. Under matched 2.2 calls per question, Qwen uses 11,106/14,018 input tokens and 3.94/13.59 seconds for Direct/RAVEL; Astra uses 11,817/15,250 tokens and 20.77/24.22 seconds. Equal ac-

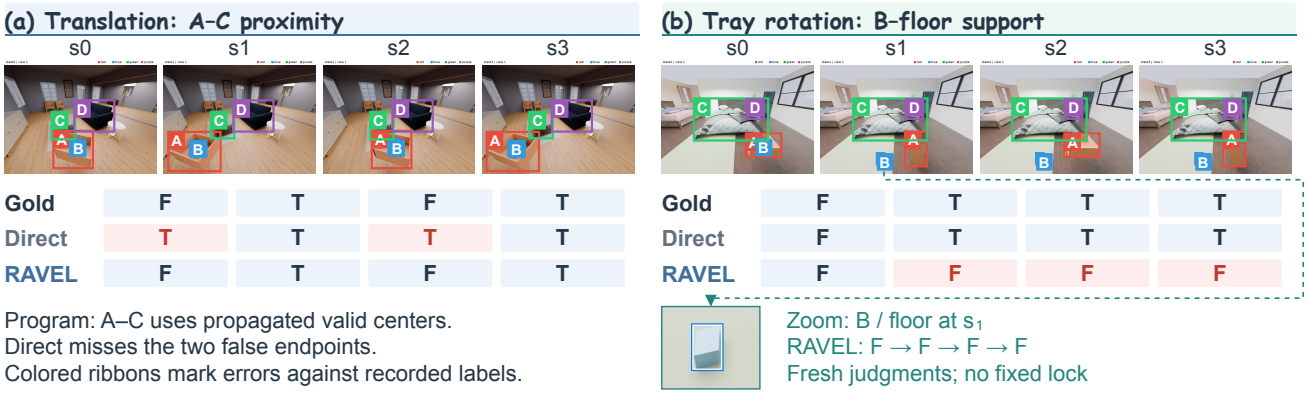


Figure 6. **Recorded correction and regression across four states.** (a) Astra RAVEL correctly alternates A–C proximity using valid geometry. (b) Its fresh visual findings miss B–floor support; Direct is correct. Red ribbons mark errors. The inset magnifies the actual state-1 image; view 1 is shown, while calls use both views. Selection is documented in Appendix E.

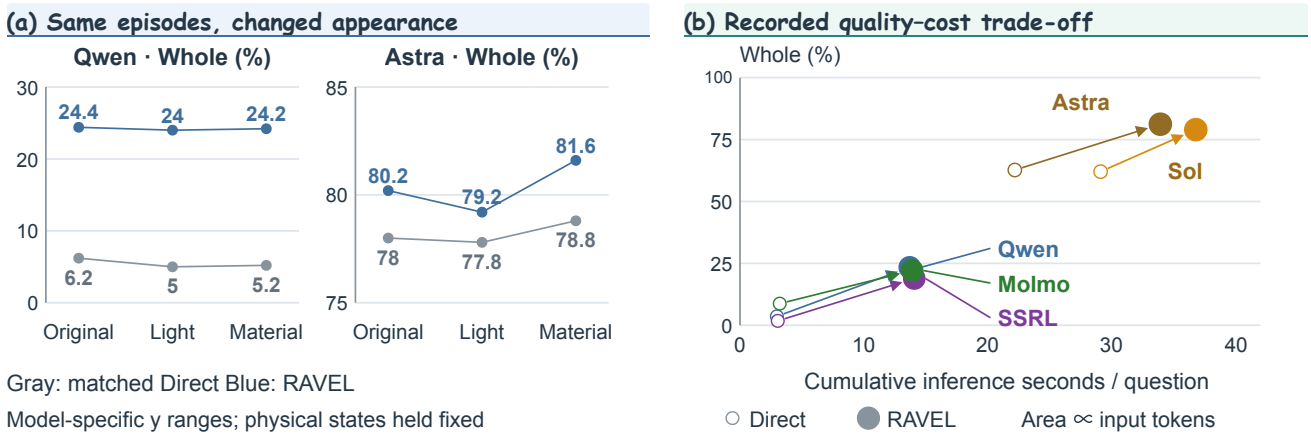


Figure 7. **Appearance robustness and recorded cost.** (a) Both matched methods use the same 500 physical episodes under original, alternative-lighting, and alternative-material renders; y-axis ranges differ by backend. (b) Five-backend full-cohort results: arrows connect Direct to RAVEL; bubble area is proportional to input tokens. Time sums calls per question; differing hardware/API conditions preclude a common hardware efficiency ranking.

cess and schedules therefore leave additional context and generation costs (Tabs. 17 and 34).

7. Limitations and Conclusion

The benchmark uses synthetic indoor scenes, four named objects, and identity boxes. Development exposed 99 of 100 source groups; independent source transfer and human audit remain pending, with two unresolved joint annotations. RAVEL requires precise initial declarations and a finite set of supported operations; fixed locks can retain initial errors. Matched controls cover two backends, and bootstrap intervals describe fixed runs without multiple-comparison adjustment. External benchmark comparisons and repeated full runs are unmeasured. The supplied kit supports offline scoring and seven complete examples; full inference reproduction additionally requires the complete images, assets, weights, and environment.

We introduce a benchmark and inference harness for joint change-and-preservation reasoning under physical actions. RAVEL models space in code: executable declarations link action-dependent attribute validity to computed, fresh, and retained findings, from which the harness composes complete relational answers. The full configuration improves all five backends; matched controls retain a large Qwen gain and a smaller Astra gain. Qwen's improvements span changes and preservation, with its largest increment in executable organization. Unresolved judgments and retained initial mistakes continue to limit complete trajectories.

LLM Usage Statement

OpenAI Codex (GPT-6-based) assisted with drafting and revising text throughout the manuscript, organizing the presentation of supplied experiments, checking related-work references, and writing LaTeX and editable figure-generation code. This assistance included substantive restructuring and compression, table assembly from supplied results, and diagram and plot layout. Scene images in the figures are display-adjusted copies or labeled crops of recorded renders; the figure-generation workflow does not synthesize experimental scenes or model responses. The vision-language models evaluated as research systems and their inference configurations are described in the experimental protocol and appendix.

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